SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
Department of Computing Technologies
Mini Project Report — Data Science | AIML-B 2026

RetailInsight

Customer Purchase Pattern Analysis Using Data Mining

Domain	Business Analytics
SDG Goal	Goal 9 — Industry, Innovation and Infrastructure
Dataset	Online Retail II (UCI / Kaggle)
GitHub Repository	MiniProject_DS_AIML-B_2026_RetailInsight
Submission Date	April 2026

Team Members	
V Shreya Team Leader	**Shaik Arbaaz** Member

1. Problem Identification

Retail businesses generate massive transactional data yet often fail to leverage it for strategic decisions. This project identifies three core analytical gaps in conventional retail reporting:

- Product Association Mining: Businesses lack visibility into which products are frequently co-purchased. Without this knowledge, cross-selling opportunities and bundle strategies remain unexploited.
- Customer Segmentation: Treating all customers uniformly leads to wasted marketing spend. Identifying distinct behavioural tiers allows targeted, high-ROI campaigns.
- Temporal Demand Patterns: Seasonal and intra-week demand fluctuations are poorly understood, leading to stockouts during peaks and overstock during troughs.

Research Objective

To apply data mining techniques — Apriori association rule mining and K-Means customer segmentation — to real-world UK e-commerce transactions, producing actionable business intelligence on purchasing patterns, customer behaviour, and demand seasonality.

2. Dataset Collection

We selected the Online Retail II dataset from the UCI Machine Learning Repository, mirrored on Kaggle. This is a real-world transactional dataset from a UK-based non-store online retailer specialising in all-occasion giftware, covering December 2009 through December 2011.

Column	Type	Description
Invoice	String	Invoice number; prefix 'C' = cancellation
StockCode	String	Unique product code
Description	String	Product name
Quantity	Integer	Units per transaction line
InvoiceDate	DateTime	Date and time of invoice
Price	Float	Unit price in GBP (£)
Customer ID	Float	Unique customer identifier (nullable)
Country	String	Country of customer residence

The dataset was placed in dataset/raw_data/online_retail_II.xlsx and loaded via pandas.read_excel() across both sheets, yielding 1,067,371 combined rows before cleaning.

3. Data Cleaning & Preprocessing

The raw dataset required significant cleaning before analysis. Steps were implemented in src/preprocessing.py and verified interactively in notebooks/preprocessing.ipynb:

- Null Removal: 243,007 rows (22.8%) lacked a Customer ID — essential for customer-level analysis. These were dropped. Rows with missing Description values were also removed.

- Cancellation Filtering: 18,744 rows with Invoice numbers prefixed 'C' representing cancellations were removed as they do not represent genuine purchase events.
- Invalid Value Removal: 71 rows with Quantity ≤ 0 or Price ≤ 0 were removed as data entry errors or system-generated adjustments.
- Column Standardisation: Column names were normalised. String columns were upper-cased and whitespace-stripped.
- Feature Engineering: Six derived features were created — TotalPrice (Quantity × Price), Year, Month, DayOfWeek, Hour, and YearMonth (YYYY-MM period string).

Stage	Rows	Removed
Raw loaded	1,067,371	—
After null removal	824,364	243,007
After cancellations	805,620	18,744
After invalid values	805,549	71
Final clean dataset	**805,549**	**261,822 total (24.5%)**

4. Exploratory Data Analysis

EDA was conducted to understand the distribution and characteristics of the cleaned dataset before applying any models. Key findings:

- Geographic Distribution: The United Kingdom accounts for ~89% of all transactions. The next largest markets are Netherlands, Germany, and EIRE (Ireland), each 1–3%.
- Revenue Concentration: Top 10 products account for a disproportionate share of revenue. REGENCY CAKESTAND 3 TIER and WHITE HANGING HEART T-LIGHT HOLDER are consistently highest-revenue SKUs.
- Temporal Patterns: Transaction volume peaks sharply in November–December, confirming holiday season demand. Thursday records the highest weekday count. Peak purchasing hours are 10:00–15:00.
- RFM Distribution Insights: Recency is right-skewed — most customers purchased within the last 100 days. Frequency heavily concentrated at 1–10 invoices. Monetary shows extreme right skew with one customer generating £608,821.

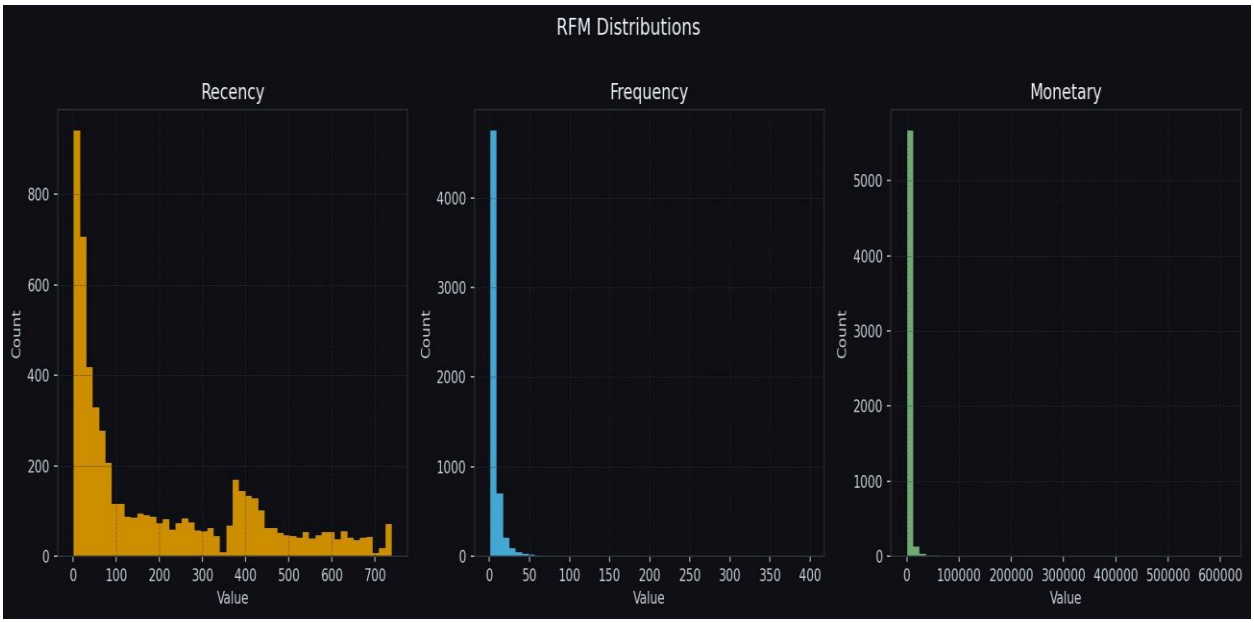

Figure 1: RFM Distributions — Recency (left), Frequency (centre), Monetary (right). All three show significant right-skew.

5. Data Visualisation

Six key visualisations were produced to communicate findings. All charts are saved to outputs/graphs/ in the repository.

#	File	Description
1	monthly_revenue_trend.png	Monthly revenue time-series across Dec 2009 – Dec 2011
2	top10_products_revenue.png	Top 10 products by total revenue — horizontal bar chart
3	transaction_heatmap.png	Transaction volume heatmap — Year × Month grid
4	day_hour_patterns.png	Revenue by Day of Week and Hour of Day
5	product_wordcloud.png	Product description word cloud — frequency-weighted
6	top_association_rules.png	Top 15 association rules by Lift (colour = Confidence)
7	support_confidence_scatter.png	Support vs Confidence scatter (bubble size = Lift)
8	rfm_distributions.png	Histograms of Recency, Frequency, and Monetary distributions
9	cluster_pca_scatter.png	PCA 2D projection of K-Means customer segments
10	segment_pie_chart.png	Customer count distribution across 4 segments
11	segment_radar.png	Normalised RFM radar profile per customer segment

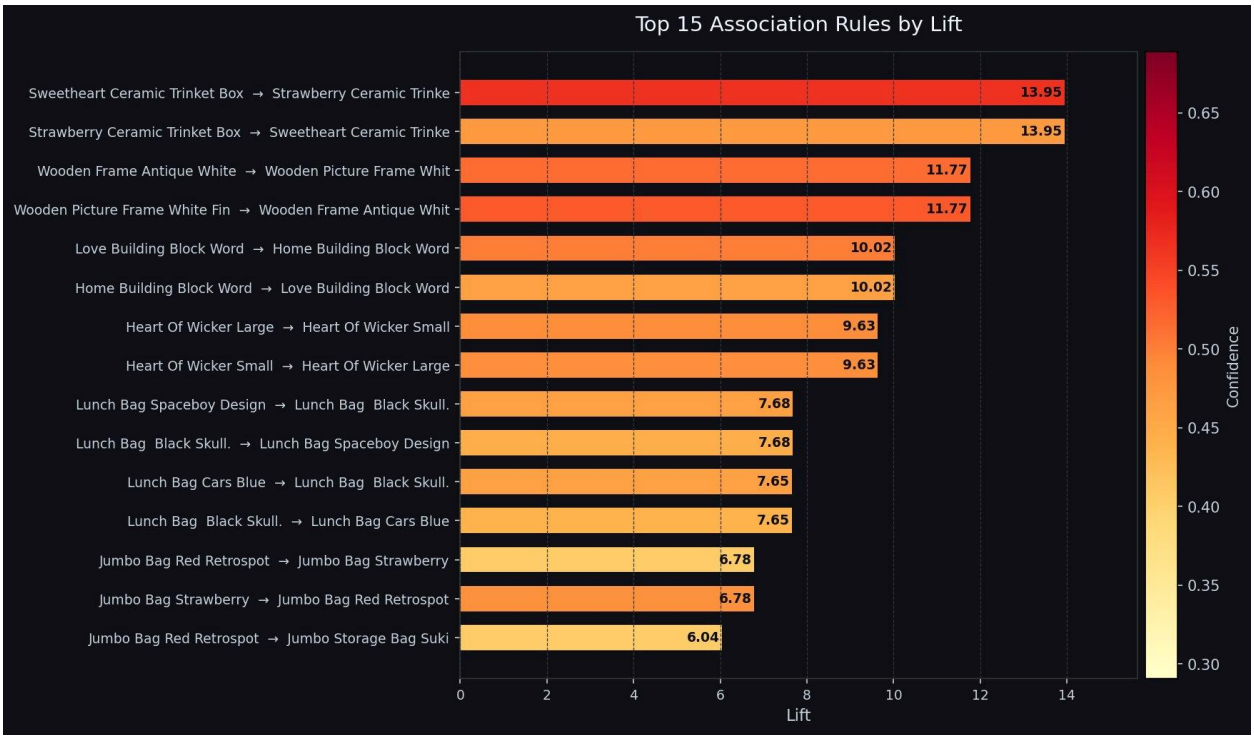

Figure 2: Top 15 Association Rules by Lift. Colour encodes Confidence (yellow = 0.30, red = 0.69).

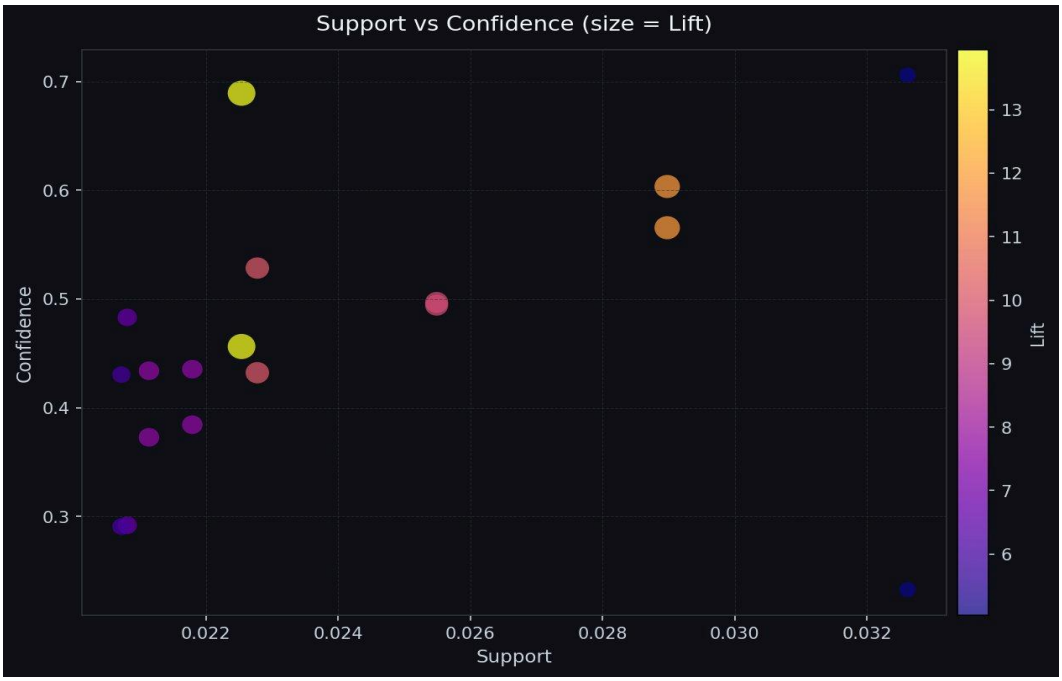

Figure 3: Support vs Confidence scatter. Bubble size encodes Lift value. Yellow bubbles = highest lift rules.

6. Model Development

Two models were developed, implemented in src/analysis.py and src/model.py respectively.

6.1 Association Rule Mining (Apriori)

The Apriori algorithm (mlxtend library) was applied to UK-market transactions only (~89% of volume). A basket matrix was constructed with invoices as rows and products as columns, boolean values indicating purchase presence.

Parameter	Value	Rationale
Market filter	United Kingdom	Largest market; prevents country-specific noise
min_support	0.02	Product pair appears in ≥ 2% of invoices
Min. appearances	10 invoices	Removes extremely rare combinations
Rule metric	Lift	Normalises for product popularity
min_threshold	1.5	Only rules stronger than random chance

Results

180 frequent itemsets discovered. 18 association rules generated, with maximum Lift of 13.95 (Sweetheart ↔ Strawberry Ceramic Trinket Box pair). The Lunch Bag category showed 3 strong internal rules with Lift > 7.6.

6.2 RFM Analysis + K-Means Clustering

RFM scores were computed per customer using a snapshot date of 10 December 2011 (one day after the last transaction). Each dimension was scored on a 1–5 quintile scale, then the combined R, F, M features were StandardScaler-normalised before K-Means clustering with k=4.

Segment	Count	Avg Recency	Avg Freq.	Avg Spend	Total Revenue
Champions	4	3.5 days	212.5×	£436,836	£1,747,343
At-Risk	35	25.9 days	103.7×	£83,086	£2,908,013
Potential Loyalists	3,841	67.0 days	7.3×	£3,009	£11,559,115
New Customers	1,998	463.0 days	2.2×	£765	£1,528,958

Figure 4: PCA 2D projection of K-Means clusters. Champions are clearly isolated as outliers in PC1.

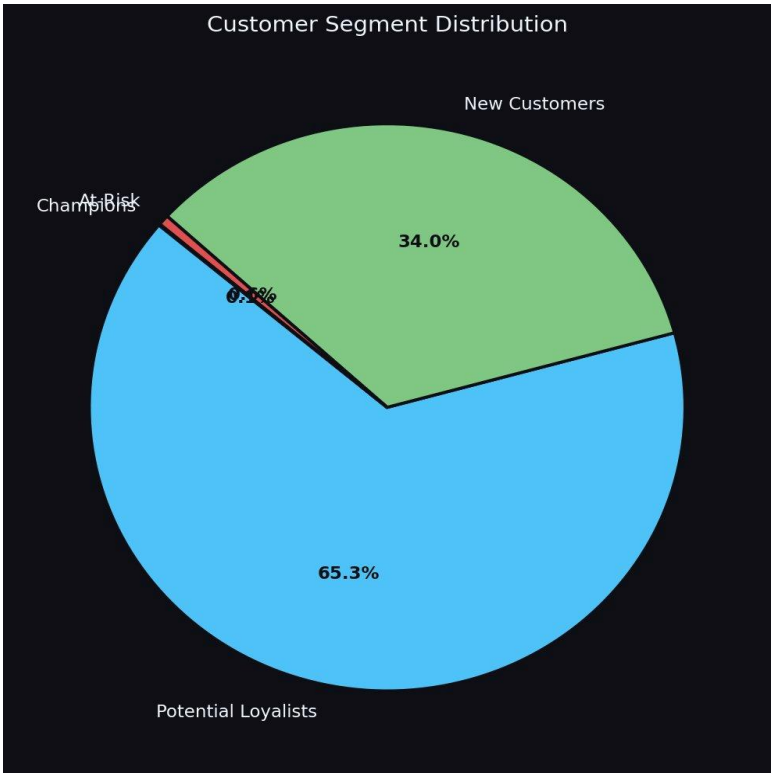

Figure 5: Customer segment distribution by count. Potential Loyalists form 65.3% of the customer base.

Figure 6: Radar chart of normalised RFM profiles per segment. Champions dominate all three axes.

7. Result Interpretation & Business Recommendations

The following findings and recommendations emerge directly from the model outputs:

01

Sweetheart ↔ Strawberry Ceramic Trinket Box (Lift: 13.95, Confidence: 69%)
Recommendation: Bundle these SKUs in product pages, recommendation emails, and checkout cross-sell prompts. A 'Complete the Set' promotion could directly capture the 69% conditional purchase probability.

02

Wooden Frame Pairs (Lift: 11.77, Confidence: 60%)
Recommendation: The Antique White and White Finish wooden frame variants are nearly always purchased together. Consider a two-pack bundle at a marginal discount to increase average order value.

03

65.3% of customers are Potential Loyalists (3,841 accounts, £11.56M revenue)
Recommendation: Highest-ROI segment for upselling. Targeted email campaigns using association rules to suggest next-purchase items can convert a significant portion toward the Champion tier within 6 months.

04

35 At-Risk customers hold £2.9M in historical revenue
Recommendation: Average spend of £83,086 per customer. Win-back campaigns with personalised offers should be prioritised within 30 days before churn becomes permanent.

05	**November–December: ~40% revenue spike (holiday season)** ***Recommendation:*** Inventory for top-selling gift items (cakestands, heart-shaped décor, bags) should be positioned by October. Pre-season supplier orders and logistics planning are critical.
06	**34% of customers are New Customers (1,998 accounts) with low engagement** ***Recommendation:*** Onboarding email sequences using association rules for personalised first-purchase follow-up recommendations can significantly improve 60-day retention rates.

8. Conclusion

RetailInsight demonstrates a complete, reproducible data science pipeline applied to real-world retail data. Starting from 1,067,371 raw transactions, the project produced 18 validated association rules (max lift 13.95), 4 statistically distinct customer segments, and 11 publication-quality visualisations — all from modular, well-documented Python source code.

The findings demonstrate that data mining can surface highly actionable business intelligence: the strongest product pairs, the highest-risk dormant customers, and the largest revenue opportunity segments — all from transaction logs alone, requiring no additional data collection.

Future Work

Potential extensions include: (1) Collaborative filtering for personalised product recommendation; (2) Time-series forecasting using ARIMA or Facebook Prophet for demand planning; (3) A real-time Streamlit or Power BI dashboard; (4) Churn prediction model using the RFM features as inputs to a classifier.

9. References

[1] Chen, D. (2015). Online Retail II Dataset. UCI Machine Learning Repository. https://archive.ics.uci.edu/ml/datasets/Online+Retail+II

[2] Agrawal, R., & Srikant, R. (1994). Fast algorithms for mining association rules. Proceedings of the 20th VLDB Conference, 487–499.

[3] Hughes, A.M. (1994). Strategic Database Marketing. Probus. [RFM Analysis origin]

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[5] Raschka, S., & Mirjalili, V. (2019). Python Machine Learning (3rd ed.). Packt Publishing.